

tree that leads to optimal decisions is a paradigmatic example of a machine learning algorithm.

Let's consider a real-world scenario in which we might use decision trees. In emergency rooms, an important decision-maker must perform triage for every newly admitted patient. *Triage* simply means assigning priority: someone who is minutes from death but can be saved by a timely operation will be admitted to treatment immediately, whereas someone who has a paper cut or a mild case of sniffles will be asked to wait until more urgent cases can be cleared up.

Triage is difficult because you have to make a reasonably accurate diagnosis with very little information or time. If a 50-year-old woman comes to the emergency room and complains of bad chest pain, the person in charge of triage has to decide whether her pain is more likely to be heartburn or a heart attack. The thought process of a person who makes triage decisions is necessarily complex. They'll take into account a number of factors: the age and sex of the patient, whether they are obese or a smoker, the symptoms they report and the way they talk about them, the expression on their face, how busy the hospital is and what other patients are waiting for treatment, and factors that they may not even be consciously aware of. In order to become good at triage, a person has to learn many patterns.

Understanding the way a triage professional makes a decision is not easy. [Figure 9-1](#) shows a hypothetical, totally made-up triage decision process (not meant as medical advice—don't try this at home!).